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Verbal Fluency as a Screening Tool for Mild Cognitive Impairment

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Abstract

Objectives: The goal of this study was to evaluate the ability of semantic (animal naming) and phonemic (FAS) fluency in their ability to discriminate between normal aging, amnestic-Mild Cognitive Impairment (a-MCI), and Alzheimer's disease (AD).

Design: We used binary logistic regressions, multinomial regressions, and discriminant analysis to evaluate the predictive value of semantic and phonemic fluency in regards to specific diagnostic classifications.

Setting: Outpatient Geriatric Neuropsychology Clinic.

Participants: 232 participants (normal aging=99, a-MCI=90, AD=43; mean age =65.75 years).

Measurements: Mini Mental Status Exam, Controlled Oral Word Association Test

Results: Results indicate that semantic and phonemic fluency were significant predictors of diagnostic classification, and semantic fluency explained a greater amount of the discriminant ability of the model.

Conclusions: These results suggest that verbal fluency, particularly semantic fluency, may be an accurate and efficient tool in screening for early dementia in time-limited medical settings.

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Descriptions of authors' role:

M. McDonnell updated, revised, coordinated submission for the manuscript. L. Dill formulated the research idea and wrote the initial manuscript draft. S. Panos assisted with statistical analyses. S. Amano and W. Brown supervised writing of manuscript. S. Giurgius assisted with data collection. G. Small was the principal investigator. K. Miller supervised data collection, statistical analyses, and manuscript preparation.

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Keywords

Dementia; Mild Cognitive Impairment; Alzheimer's disease; screening

Introduction

The task of screening for and diagnosing dementia predominantly occurs in primary care settings, where accurate detection is dependent upon a patient's symptom severity (Boustani et al., 2005; Lopponen et al., 2003; Mitchell et al., 2011; Parmar et al., 2014). As a result, patients early in the disease process are often less accurately diagnosed (Mitchell et al., 2011; Parmar et al., 2014). This is of concern as research has shown that early diagnosis and intervention of Alzheimer's disease (AD) has important clinical and economic implications. Early intervention allows for increased preservation of cognition and improved daily functioning, which improves quality of life and reduces the financial burden of long-term placement (Leifer 2003).

Diagnostic accuracy in primary care is dependent upon the primary care provider's (PCP) knowledge of aging and cognitive functioning. Individuals with normal age-related decline, also known as age-associated memory impairment (AAMI), have memory performances that are within normal limits compared to their same-aged peers, but fall at least one standard deviation below that of young adults (Crook et al., 1986). By contrast, mild cognitive impairment (MCI) is often considered a transition stage between normal aging and dementia (Albert et al., 2011; Moorhouse, 2009; Petersen et al., 2001). The different subtypes of MCI include amnestic-MCI (a-MCI), which involves memory impairment, and MCI-nonamnestic (n-MCI), which involves impairment in cognitive domains other than memory. Research has shown that subjects diagnosed with a-MCI are at a significant increased risk for converting to AD (Albert et al., 2011; Fischer et al., 2007; Petersen et al., 2001). Neuroimaging studies indicate that patients with a-MCI who convert to AD have less grey matter volume in the brain's temporal and parietal lobes (Karas et al., 2008). Additionally, postmortem studies have shown that patients with a-MCI have neuropathological changes in the brain that are characteristic of AD (Petersen et al., 2001).

As research has demonstrated the importance of early detection and intervention of AD, cognitive screening tools are becoming an essential part of this detection process. The Mini Mental Status Exam (MMSE; Folstein et al., 1975) is the most widely used cognitive screening tool in part because of its relatively short administration time (5–10 minutes) and its ease of administration and scoring (Strauss et al., 2006). Studies have shown that the MMSE is a reliable screening tool for detecting dementia (Boustani et al., 2003; Folstein et al., 1975; Jackson and Ransdell, 1988; Moorhouse 2009), but is less sensitive in detecting a-MCI (Lehrner et al., 2005; Nasreddine et al., 2005; Tang-Wai et al., 2003). As a result, the Montreal Cognitive Assessment Test (MoCA) has become an increasingly popular screening tool as it is able to detect early stages of cognitive impairment (Nasreddine et al., 2003). While this instrument has been shown to have greater sensitivity than the MMSE, it takes an additional ten minutes for physicians to administer, which is not practical in many primary care settings. Moorhouse in 2009 reported that most PCPs spend roughly 11

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minutes consulting with each patient, thus require a screening test that could be completed quickly (Moorhouse 2009). As a result, researchers have proposed an additional screening measure to the MMSE, specifically verbal fluency, to aid in detection of mild cognitive impairment while maintaining the brevity of the measure (Galasko et al., 1990; Sebaldt et al., 2009).

Consensus has been reached that both semantic and phonemic fluency tasks are impaired in neurodegenerative disorders such as AD (Fama et al., 2000; Hodges et al., 1992), with greater impairment in semantic relative to phonemic fluency tasks in the AD population (Chan et al., Henry et al., 2004; Monsch et al., 1992). Similarly, individuals with a-MCI have shown comparable deficits on semantic fluency tasks relative to phonemic tasks. Individuals with a-MCI also have been shown to perform less than their peers on both verbal fluency tasks, despite maintaining scores within the low average range (Nutter-Upham et al., 2008; Rinehardt et al., 2014). Their performance is typically characterized by fewer semantic subcategories and words compared to healthy individuals (Price et al., 2012). These results suggest that verbal fluency measures may be sensitive to more subtle declines found in a-MCI. In contrast, normal aging individuals typically have robust performances on verbal fluency tasks, including phonemic and semantic fluency. Normative data indicates that for individuals experiencing normal aging, individuals between the age of 50–70 will typically recall 21 animals, and individuals aged 70 and older recall approximately 18 animals (Price et al., 2012), which is higher than individuals with a-MCI or AD, who were able to list 15 and 8 animals, respectively (Clark et al., 2009). With regard to phonemic fluency, normal aging individuals are able to recite 41 words per minute, while those with a-MCI and AD recall 37 and 20 words per minute, respectively (Clark et al., 2009). This suggests that verbal fluency typically remains intact during the normal aging process (i.e., no signs of a neurodegenerative disease present).

While current literature supports the relationship between verbal fluency decline, particularly semantic fluency, in both pre-clinical AD (a-MCI) and clinical AD, very few studies have looked at using verbal fluency measures as brief cognitive screening tools for a-MCI. Of those studies, results suggested that animal fluency showed similar strengths in specificity, sensitivity and accuracy to the MoCA in the detection of cognitive impairment in a group of older adults with vascular risk factors (Sebaldt et al., 2009). Galasko et al. (1990) found that adding a verbal fluency measure (FAS) to the MMSE improved the accuracy of diagnosis of patients with AD who scored in the non-demented range on the MMSE.

In light of these findings, we examined semantic and phonemic fluency collectively and individually to determine the strongest predictive value, while also assessing the effects of age, education and gender. Given the literature that supports semantic fluency as an accurate predictor of pre-clinical and clinical AD, it is hypothesized that semantic fluency will have greater ability to discriminate between normal aging, a-MCI, and AD than phonemic fluency. We also hypothesized that combining semantic fluency and phonemic fluency will add greater predictive value than semantic or phonemic fluency alone.

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Methods

Participants

Participants were retrospectively identified from archival data collected in the Aging and Memory Research Center at the UCLA Semel Institute for Neuroscience and Human Behavior. Recruitment through advertisement and physician referral emphasized middle-aged and older people with memory complaints. Any subjects with a neurological, medical, or psychiatric condition that could affect memory or other cognitive processes were excluded. Written informed consent was obtained in accordance with the procedures set by the UCLA Institutional Review Board. Subjects were part of a longitudinal study wherein they completed comprehensive neuropsychological evaluations at baseline, as well as a 2-year (mean [M)]=2.4; standard deviation [SD]=1.3 years) follow-up re-evaluation. For this particular study, comprehensive neuropsychological data collected at baseline was analyzed to evaluate predictive value of the MMSE, phonemic and semantic verbal fluency on ability to discriminate between normal aging, a-MCI, and AD at that moment.

This sample consisted of 91 men and 141 women, with an average age of 66 years (M=65.75, SD=11.55), and average education of 16 years (M=16.21, SD=2.77). Caucasians were the largest ethnic group (n=199), followed by Asian American (n=15), African American (n=9), Latino (n=7), and other (n=2).

Procedures

Diagnostic Groups—All participants completed a comprehensive neuropsychological evaluation, comprised of tests that are widely used in research on both normal aging and neurodegenerative disorders. These measures were used to classify subjects into diagnostic groups. Participants were divided into three diagnostic groups based on their testing performance. The normal aging group, individuals meeting criteria for AAMI, performed within normal limits (> 16th percentile) for memory tasks when compared to same age peers, though one standard deviation below the mean of healthy young adults on at least one of the six memory tasks (i.e., age associated memory impairment - AAMI). Criteria for a-MCI included: 1) Self-reported memory impairment and/or a CDR score of .5 or higher; 2) Impairment on two or more of the six delayed memory tasks (< −1 standard deviation for these 2 or more memory tasks). Notably, the normal aging and a-MCI groups all had a MMSE score of > 24, but MMSE was not used to determine classification. The AD participants were classified as meeting criteria for dementia based on 1) Impaired scores (< −1.5 standard deviation) for multiple cognitive domains (memory + one or more non-memory domains); 2) Confirmatory diagnosis from physician and/or MRI data collected as part of the research study. Note, the majority (74%) classified as AD had MMSE scores < 24, but as mentioned earlier, MMSE was not used for the specification classification into the AD category. The following cell sizes were obtained, for a total sample size of 232: normal aging (n=99), a-MCI (n=90), and AD (n=43).

Measures—In this study, the following measures were included in the analysis: the MMSE (Folstein et al., 1975), the Animal Naming Test (Newcombe 1969), and the Controlled Oral

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Word Association Test: FAS (Benton and Hamsher, 1976). Raw scores were converted to z-scores using the overall sample mean and standard deviation.

Analyses—The first set of analyses used logistic regression to assess the additive contribution of verbal fluency in classifying two groups, normal aging and cognitively impaired (a-MCI and AD). Using logistic regression, demographics were placed in the first step, MMSE in the second step, and verbal fluency measures in the final step (individually and together). In the second set of analyses, multinomial logistic regression was used to classify the three-group model (normal aging, a-MCI and AD) using the same steps above. Finally, discriminant function analysis was used as an additional follow-up analysis using demographics, MMSE, and both verbal fluency measures. Data were evaluated to ensure assumptions were met. Transformations were computed on age for logistic regression models, and the MMSE for the discriminant function analysis (due to separate assumptions). Models were also run with and without outliers and overall findings remained unchanged. SPSS 24 statistical package was used.

Results

Group differences

Descriptive statistics (for both the two-group and three-group models) are provided in Table 1. Individual test scores for FAS, Animals, and the MMSE were transformed into z-scores based on larger sample mean and standard deviation, controlling for age, gender, ethnicity, and level of education. Among the demographic variables, age was a significant predictor in both the two-group and three group models ($p < .001$). In both the two group and three-group models, there was a significant difference in all three variables of interest (MMSE, FAS, Animals) in the expected direction, with the cognitively impaired group performing worse than normal aging group (*p's* < .001). In the three-group model, follow-up multiple comparison analyses indicated that the AD group performed lower than the a-MCI and normal aging (AAMI) groups (*p's* <.001), and the a-MCI group performed lower than the AAMI group in all three variables (*p's* < .001).

Using logistic regression, demographics were added in the first step, MMSE into the second step, and verbal fluency measures were added in the third steps (individually and both together). See Table 2 for overall fit statistics. As seen in Table 2, after controlling for demographics, the MMSE demonstrated 70.7% sensitivity and 71.7% specificity (Overall Percentage Correct = 71.1%). The addition of FAS improved the overall predictive validity of the model (χ^2 [1, N = 232] = 22.9, $p < .001$), by improving sensitivity (77.4%) though somewhat at the cost of specificity (67.7%) (Overall Percentage Correct = 73.3%). Animals also improved the overall predictive validity of the model when added to the MMSE (χ^2 [1, N = 232] = 61.5, $p <.001$), by improving both sensitivity (81.2%) and specificity (77.8%) (Overall Percentage Correct = 80.6%). In a separate analysis, assessing the contribution of animals above and beyond FAS, the addition of animals evidenced an additive effect (χ^2 [1, N = 232] = 41.88, $p < .001$), by improving both sensitivity (83.5%) and specificity (77.8%) (Overall Percentage Correct = 81.0%). In this final model, with all variables combined

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animals remained significant ($p < .001$), though FAS was just below significance ($p = .06$). Please see Table 3 for coefficients of this final model.

In the next set of analyses, multinomial regressions were used to determine whether verbal fluency would aid in predicting the three-group model in a similar manner. Overall, the findings followed largely the same pattern. See Table 4 for fit statistics. Both FAS (χ^2 [2, N = 232) = 25.70, p <.001) and Animals (χ^2 [2, N = 232] = 63.43, p <.001) added predictive validity to the models. When all variables were combined (demographics, MMSE, and both fluency measures), in addition to animals, age (χ^2 [2, N = 232) = 10.35, p = .01), and MMSE (χ^2 [2, N = 232) = 76.78, p <.001) remained significant predictors though FAS fell below significance χ^2 [2, N = 232) = 4.76, p = 09). Please see Table 5 for the individual parameter estimates. Overall, in this final three-group model, age (p<.01) and animals (p<.01) were significant predictors of group membership between the healthy aging group (AAMI) and the a-MCI group and MMSE and animals were significant predictors of AD group membership when compared to the healthy aging group.

Finally, a discriminant function analysis using all six predictor variables revealed two discriminant functions. The first function explained 92.7% of the variance, whereas the second explained 7.3%. In combination, these discriminant functions significantly differentiated the diagnostic groups, with a canonical correlation of .82, Λ=.29, χ^2(12, N=232) =279.61, p <.001, and by removing the first function indicated that the second function also significantly discriminated the diagnostic groups, with a canonical correlation of .37, Λ=.37, χ^2(5, N=232) =32.86, p<.001. The overall model correctly categorized 78.4% of the groups (79.8% of the healthy aging; 74.4% of the aMCI; and 83.7% of AD groups).

Discussion

The goal of this study was to determine whether semantic or phonemic fluency would be a useful addition to the MMSE in detecting a-MCI or pre-clinical AD. Verbal fluency scores were assessed to determine their ability to differentiate among normal aging (AAMI), a-MCI, and AD groups. It was hypothesized that semantic fluency would have greater ability to discriminate between the three diagnostic groups than phonemic fluency, while combining semantic and phonemic fluency would add greater predictive value than semantic or phonemic fluency alone. Results supported the initial hypothesis that semantic fluency would add significant predictive value to the MMSE, however, the hypothesis that the combination of phonemic and semantic fluency would add even greater predictive value was not supported by these analyses.

Group comparisons demonstrated significant group differences in performance for both semantic and phonemic fluency, with the a-MCI group performing in the average and low average range for FAS and Animals, compared to the normal aging group (AAMI) that performed in the average and high average range, respectively. These results, along with our mean scores for the a-MCI group, are similar to a study by Murphy, Rich, and Troyer (2006), which examined verbal fluency patterns in a-MCI compared to normal controls and AD subjects (Murphy et al., 2006). They also found the a-MCI group to demonstrate a

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decline in semantic fluency relative to phonemic fluency compared to controls, despite the fact that their overall performance fell within the normal range of ability.

We also found that while the MMSE and age had adequate discriminability in the two group models, the addition of fluency measures substantially improved prediction, with FAS improving sensitivity, and the addition of animals improving both sensitivity and specificity compared to demographic factors and the MMSE. In the three group models, MMSE was unable to detect a-MCI well, and the addition of fluency measures, especially animals, improved sensitivity and specificity. When determining the effect of each of the variable combined, results further supported that Animals improved the predictive value above beyond the contribution of FAS. As the MMSE is the most widely used cognitive screening tool, this further suggests that a verbal fluency measure, particularly a semantic fluency measure, would be a helpful addition as it demonstrates almost equal importance in the predictive model as well as excellent sensitivity. Of note, the 95% Confidence Intervals for sensitivity and specificity appear to overlap, suggesting there is no statistically significant difference with the addition of Animals and FAS. However, in many cases, despite an overlap in confidence intervals, results may remain statistically significant (Austin and Hux, 2002), and given the p-values ($p < .001$) across models, it suggests that this data remains statistically significant despite the overlap in confidence internals. This is also similar to the findings of Galasko et al. (1990), who found that adding a verbal fluency measure improved the diagnosis of patients with AD who were previously undetected by the MMSE alone.

The findings of this study may have important implications for the screening and detection of pre-clinical AD. From a practical standpoint, Animal naming compared to FAS would be a quick (1-minute administration time). It is also a relatively easy test to score and interpret, requiring little skill in administration. Although alternative cognitive screens for a-MCI have been created, such as the MoCA, these are typically much longer to administer and are not nearly as widely used by physicians, such as neurologists, psychiatrists, and family practitioners. Therefore, the incorporation of semantic fluency tasks (Animal naming) could possibly be an efficient way to handle the deficiencies of the MMSE with marginal time loss or error. Additionally, research supports the link between verbal fluency and the gradual progression associated with AD. Both semantic and phonemic fluency has been found to be reduced, with semantic fluency being significantly lower than phonemic fluency in those at risk for developing AD (Salmon et al., 1999). Furthermore, as the disease progresses, while performances on both tasks decline, the rate of decline associated with semantic fluency is faster than that of phonemic fluency. This pattern of significant decline in semantic fluency is also apparent when evaluating the difference between normal controls and those with a-MCI, despite the fact that a-MCI patients still performed significantly better than those diagnosed with AD in both semantic and phonemic fluency (Murphy et al., 2006).

Limitations of this study involve homogeneity and methodology. First, data were collected in a largely urban, highly educated, Caucasian setting. While our findings show education to have had no significant effect on verbal fluency performance, it is possible that the outcome could differ with a less homogenous group. In addition, as the a-MCI group was relatively young (m=64.77) compared to similar studies (m=71.76, sd=6.66 [Nutter-Upham et al., 2008]; m=76.6, sd=5.4 [Murphy et al., 2006]), the study was rerun, excluding the

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youngest participants (<60 years of age) to assess possible age biases. As the results yielded no significant differences from the full and trimmed models, the younger participants were not excluded from the results. Finally, potential limitations also exist in the selection and conduction of the data analysis. While the inclusion of the predictor variables in the discriminant function analyses was based on extensive literature review, the trimming of the model was determined statistically and at the discretion of the researcher. As a result, the possibility exists that unique features of the data set were influential upon the final models, and therefore should be interpreted with relative caution, particularly in light of the study's lack of replicative data.

Future studies would benefit from comparing semantic fluency to other sensitive measures to find the most sensitive and specific cognitive screening combination. Due to aforementioned weaknesses in this study, further study is needed to develop a range of cut-off scores for Animal naming. This way, physicians would be able to interpret results based on cutoff scores, aiding implementation and reducing error. Additional research would also benefit from evaluating the performances of patients with other variations of cognitive deficits (e.g., MCI due to vascular dementia, dysexecutive-MCI) to establish diagnostic sensitivity across diagnosis types. We hope that this study will be the catalyst for future studies in this area, as verbal fluency may offer a well-needed solution to the cognitive screening conducted in our time constrained medical profession.

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Conflicts of Interest Declaration

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Table 1.

Demographics of Two and Three Diagnostic Groups

	Two-Diagnostic Group Model			Three-Diagnostic Group Model				
	Normal Aging (AAMI)	**Cognitive Impairment (a-MCI + AD)**		**Normal Aging (AAMI)**	**a-MCI**	**AD**		
	(*n*=99)	**(*n*=133)**		**(*n*=99)**	**(*n*=90)**	**(*n*=43)**		
	M (SD)	***M (SD)***	***p***	***M (SD)***	***M (SD)***	***M (SD)***	***p***	**Bonferroni Correction**
Age	63.48 (11.73)	67.43 (11.16)	0.01	63.48 (11.73)	64.77 (11.1)	73.00 (9.02)	<.01	AAMI < AD; a-MCI <AD
Sex (% Male)	36.40%	41.40%	0.44	36%	46%	33%	0.27	
Education	16.30 (2.48)	15.88 (2.47)	0.20	16.30 (2.48)	16.06 (2.45)	15.51 (2.51)	0.23	
Ethnicity (%Caucasian)	88.90%	83.50%	0.24	89%	80%	91%	0.13	
MMSE	29.03 (1.19)	25.53 (4.28)	<.001	29.03 (1.19)	27.96 (1.69)	20.44 (3.53)	<.001	AAMI > a-MCI > AD
FAS	47.74 (11.45)	33.52 (12.42)	<.001	47.74 (11.45)	37.30 (10.9)	25.60 (11.7)	<.001	AAMI > a-MCI > AD
Animals	22.43 (5.08)	13.54 (5.85)	<.001	22.43 (5.08)	15.82 (5.25)	8.77 (3.81)	<.001	AAMI > a-MCI > AD

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Table 2.

Two Group Model Fit Statistics

		−2 Log Likelihood	AIC	BIC	χ^2	DF	p	Sens	95% CI Sen	Spec	95%CI Spec	Correct
Model 1												
Step 1	Demo	304.4	314.9	316.2	12.3	4	0.02	79.7	71.7–86.2	33.3	24.2–43.5	59.9
Step 2	MMSE	239.8	250.7	253.9	64.5	1	<.001	70.7	62.2–78.3	71.7	61.8–80.3	71.1
Step 3	FAS	216.9	229.1	233.5	22.9	1	<.001	77.4	63.4–84.2	67.7	57.5–76.7	73.3
Model 2												
Step 1	Demo	304.4	314.9	316.2	12.3	4	0.02	79.7	71.7–86.2	33.3	24.2–43.5	59.9
Step 2	MMSE	239.8	250.7	253.9	64.5	1	<.001	70.7	62.2–78.3	71.7	61.8–80.3	71.1
Step 3	Animals	178.3	191.2	194.8	61.5	1	<.001	81.2	73.5–87.5	78.8	70.5–87.2	80.6
Model 3												
Step 1	Demo	304.4	314.9	316.2	12.3	4	0.02	79.7	71.7–86.2	33.3	24.2–43.5	59.9
Step 2	MMSE	239.8	250.7	253.9	64.5	1	<.001	70.7	62.2–78.3	71.7	61.8–80.3	71.1
Step 3	FAS	216.9	229.1	233.5	22.9	1	<.001	77.4	63.4–84.2	67.7	57.5–76.7	73.3
Step 4	Animals	174.50	189.2	193.42	42.41	1	<.001	83.5	76.0–89.3	77.8	68.3–85.5	81.0

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Table 3.

Two-Group Model 3, Step 4 Coefficients

	Wald	DF	Sig.	Exp(B)	95% CI	
					Lower	Upper
Age[b]	9.32	1	<.001	0.94	0.90	0.98
Educ	0.09	1	0.76	0.98	0.84	1.14
Sex	0.40	1	0.53	0.77	0.35	1.73
Ethnicity	0.00	1	1.00	1.00	0.35	2.85
MMSE	6.99	1	0.01	0.28	0.11	0.72
FAS	3.59	1	0.06	0.61	0.36	1.02
Animals	29.52	1	<.001	0.12	0.06	0.26
Constant	7.93	1	0.01	263.84		

[a] All continuous variables z scored

[b] transformed variable (z score for square root)

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Table 4.

Three Group Multinomial Model Fit Statistics

		−2 Log Likelihood	AIC	BIC	χ^2	DF	p	Percent Correct	AAMI	a-MCI	AD
Model 1											
Step 1	Demo	410.13	426.13	453.70	29.06	6	<.001				
Step2	MMSE	275.74	295.74	330.21	172.29	2	<.001	68%	75%	50%	91%
Step 3	FAS	256.97	280.97	322.33	25.70	2	<.001	71%	70%	63%	88%
Model 2											
Step 1	Demo	410.13	426.13	453.70	29.06	6	<.001				
Step2	MMSE	275.74	295.74	330.21	172.29	2	<.001	68%	75%	50%	91%
Step 3	Animals	219.25	243.25	284.61	63.43	2	<.001	77%	80%	67%	91%
Model 3											
Step 1	Demo	410.13	426.13	453.70	29.06	6	<.001				
Step2	MMSE	275.74	295.74	330.21	172.29	2	<.001	68%	75%	50%	91%
Step 3	FAS	256.97	280.97	322.33	25.70	2	<.001	71%	70%	63%	88%
Step 4	Animals	214.47	242.47	290.72	42.50	2	<.001	78%	80%	71%	86%

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Table 5.

Three-Group Model 3, Step 4 Coefficients

		B	SE	Wald	df	p	Exp(B)	95% CI LB	Exp(B) UB
aMCI	Intercept	0.69	0.34	4.05	1	0.04			
	Age	−0.71	0.23	9.34	1	<.001	0.49	0.31	0.78
	Education	−0.06	0.20	0.10	1	0.75	0.94	0.64	1.38
	Sex	0.29	0.41	0.49	1	0.48	1.33	0.60	2.98
	MMSE	−0.98	0.53	3.40	1	0.07	0.38	0.13	1.06
	FAS	−0.50	0.26	3.70	1	0.06	0.61	0.37	1.01
	Animals	−2.06	0.38	28.71	1	<.001	0.13	0.06	0.27
AD	Intercept	−3.60	1.22	8.69	1	<.001			
	Age	−0.64	0.58	1.19	1	0.28	0.53	0.17	1.66
	Education	−0.01	0.46	0.00	1	0.99	0.99	0.40	2.46
	Sex	1.01	1.10	0.85	1	0.36	2.74	0.32	23.51
	MMSE	−6.22	1.39	19.91	1	<.001	0.00	0.00	0.03
	FAS	−1.03	0.67	2.38	1	0.12	0.36	0.10	1.32
	Animals	−2.69	0.86	9.77	1	<.001	0.07	0.01	0.37

Reference group is Normal (AAMI)